Magnetic Holography

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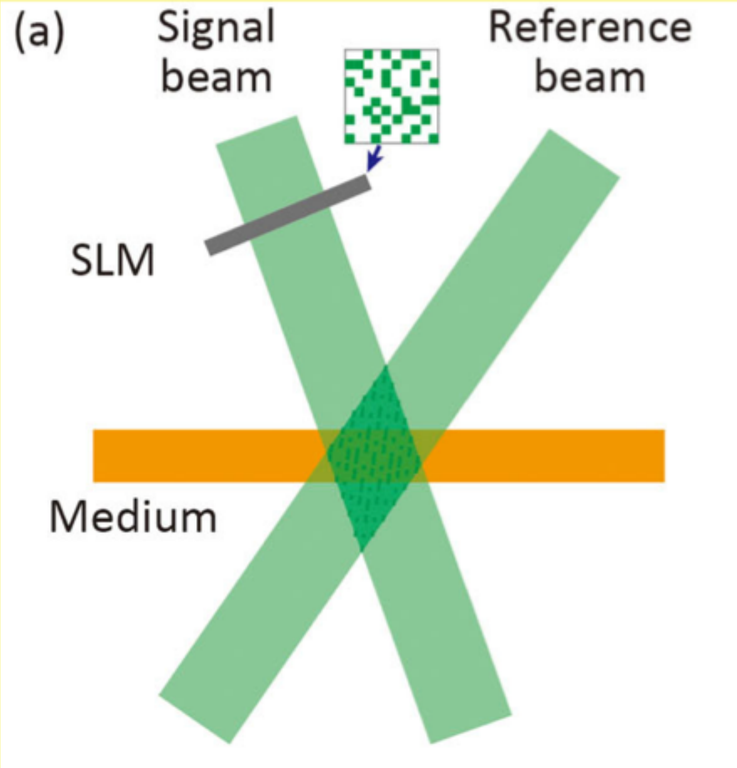
What is holography?

- Store 2D data using interference and diffraction of light
- Signal beam contains data we want to store
- Reference beam is used to create an interference pattern
- The interference pattern is recorded on some medium
- Illuminating the interference pattern with the reference beam reproduces the signal beam
- Allows for dense data storage with fast recovery

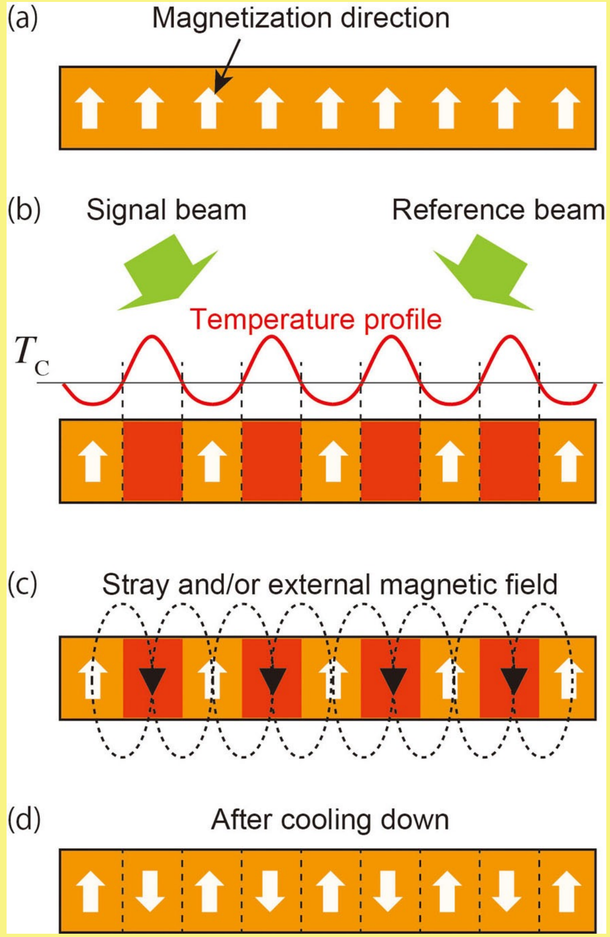


Traditional light holography

- Interference pattern is recorded as a change in refractive index of the medium
- Bright image reproduction is possible
- Problems?
- Volume can change as photo-reaction occurs
- Recordings can be unstable
- Difficult to find a rewritable medium

How magnetic holography differs

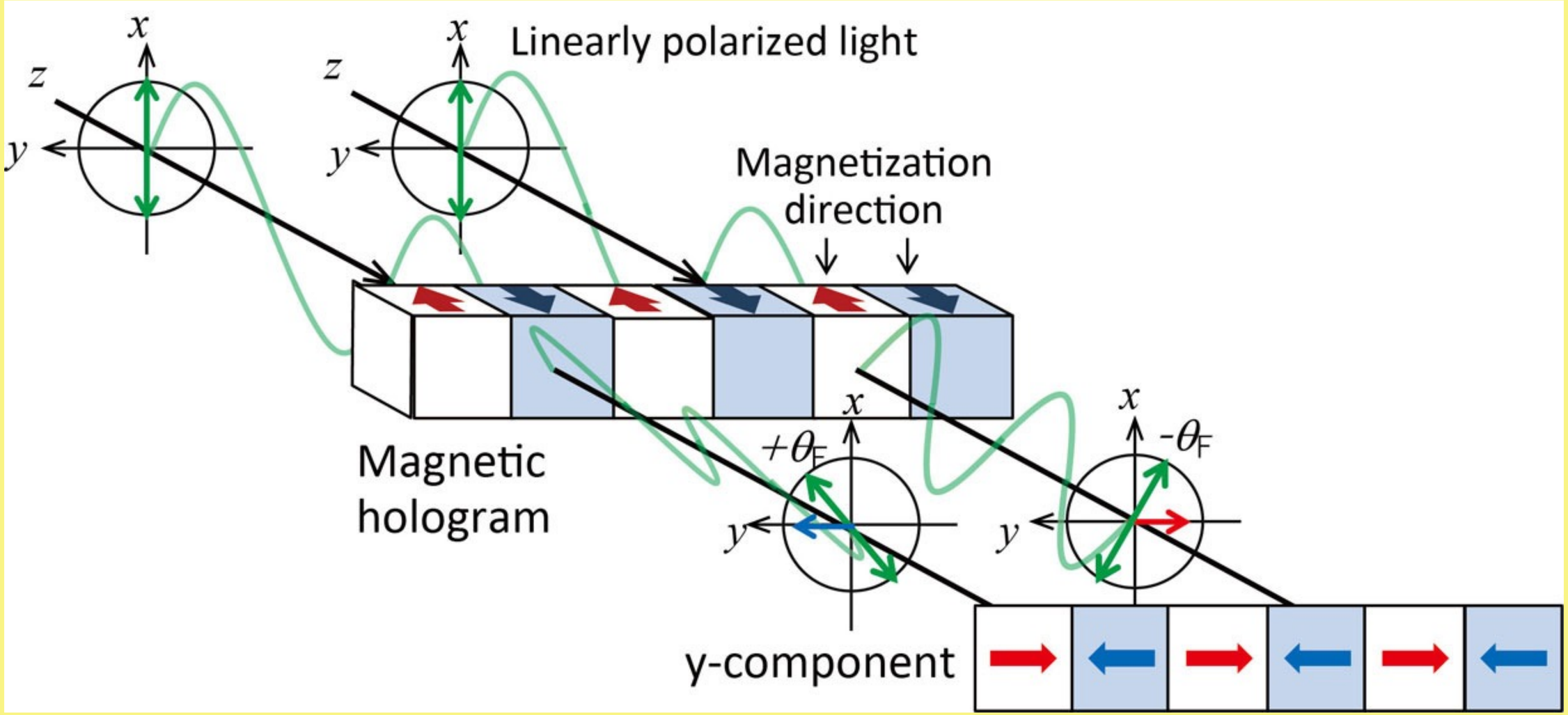
- The interference pattern heats up parts of the medium
- After a certain temperature the material loses its natural magnetisation
- An external field reverses the magnetisation direction as the medium cools
- The data is stored as a difference in magnetisation direction



Signal reconstruction

- Faraday rotation
- Magneto-optical effect causes changes in polarisation direction of light
- This indicates the direction of magnetisation at that point
- The hologram diffracts the incident reference beam
- This diffraction recreates the signal beam

Signal reconstruction



Why magnetic?

- Rewritable - The medium can be heated to reset the magnetic field
- Stable - The materials used (metal film or iron garnets) can be quite tough
- Dense data storage - Holograms can be offset by tens of microns, overlapping, and still be recoverable
- Cheap - Materials needed are either common or easy to produce

References

- [1] Y. Nakamura, “Magnetic Holography and Its Application to Data Storage,” Photonics, vol. 8, no. 6, p. 187, May 2021 [Online].
- Available: http://dx.doi.org/10.3390/photonics8060187

Thank you!

- Questions?